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STUDIES IN THE SIERRA.

NO. II.—MOUNTAIN SCULPTURE.—ORIGIN OF YOSEMITE VALLEYS.

ALL the valleys and cañons of the western flank of the Sierra, between 36° and 39° north latitude, naturally classify themselves under two genera, each containing two species. One genus comprehends all the slate valleys; the other all that are built of granite. The latter is far the more important, both on account of the greater extent of its geographical range and the grandeur and simplicity of its phenomena. All the valleys of both genera are valleys of erosion. Their chief distinguishing characteristics may be seen in the following descriptions:

SLATE VALLEYS.

1. Cross-sections, V-shaped, or somewhat rounded at bottom, *walls irregular in structure*, shattered and weak in appearance, because of the development of slaty cleavage planes and joints, which also prevent the formation of plane-faced precipices. Bottom showing the naked bed-rock, or covered by rocky *débris*,

and sloping in the direction of the trend. Nearly all of the foot-hill valleys belong to this species. Some of the older specimens are smoothly covered with soil, but *meadows and lakes are always wanting*.

2. More or less widened, *branching at the head*. Bottom, with meadows, or groves, or lakelets, or all together. Sections and walls about as in No. 1. Fine examples of this species occur on the head-waters of the San Joaquin.

GRANITE VALLEYS.

1. Cross-sections narrowly or widely V-shaped. Walls seldom interrupted by side-cañons, magnificently simple in structure and general surface character, and presenting plane precipices in great abundance. Bottom sloping in the direction of the trend, mostly bare, or covered with unstratified glacial and avalanche boulders. Groves and meadows wanting.

2. *Branching at head, with beveled and heavily abraded lips at foot*. Bot-

write me that the Indians, who had been at first exceedingly frightened, had begun to be re-assured, and that they continued to render all the services in their power to the inhabitants who remained in that quarter awaiting the new order.

You see, my reverend father, a very long letter, and perhaps one a little too long. I should esteem myself happy if it is able to afford you any pleasure, for I had no other object in writing. I am, with respect, in the unity of your holy sacrifices, etc.

THE LORDS OF THE ISLES.

AS we survey the Californian islands, the question rises to the lips: Was ever land so wasted and desolate enshrined in such a rich and sunny sea, or blessed with a climate more delightful? The delicate haze peculiar to the coast softens and enriches the dry brown color of the hills; and the harshness of the bluffs and ragged peaks is modified by the blue, calm water which ripples on the shore. At a slight distance from some of the larger islands, the tangled chaparral and weedy growth, which nestle in the deep ravines, lend a slight verdure to the landscape, and on the whole the prospect is not unpleasing.

On landing, this deceitful glamour is soon dispelled. The vegetation resolves itself into stunted, thorny bushes, prickly cactus, and sparse and unattractive weeds. Water is almost absent, and when found, except in one place on Catalina, is mawkishly sweet with alkali, or bitter with other minerals.

Clemente, San Nicolas, and, for the most part, San Miguel, are barren wastes of sand. Except the last, and including Santa Barbara and Anacapa, water is totally absent, except during the winter rains. Santa Rosa, Santa Cruz, and Catalina are more favored, but only relatively. The last mentioned five are high and rocky, and the others comparatively low.

Here and there are tolerable anchorages, and there is one good, but very

small, harbor on Catalina. The only value of the islands is for sheep ranges, though in those places where water exists, in favorable seasons, the cultivator may obtain tolerable crops, in patches of small extent. When the season is unfavorable, the sheep and cattle die by thousands, not from thirst, but from the entire desiccation and disappearance of the scanty herbage necessary to sustain them. Small land-birds are scarce, and the more plentiful quail are very shy. A diminutive fox obtains a living in some unexplained way, and occasionally a cotton-tail rabbit may be seen disappearing in a clump of cactus, with a reckless disregard of the omnipresent spines which is little short of miraculous. Beneath every stone nestle the greenish-yellow centipede, a small kind of scorpion, or the disgusting tarantula. In the summer, the common rattlesnake abounds in every bunch of shrubbery and crevice of the rocks. On the sandy surface of the lower islands may be seen, everywhere, anchored by their fore-legs and head deeply buried in the sand, a variety of beetles of the most nauseous odor and unprepossessing appearance. The large, but harmless, yellow mole-cricket which abounds there is hardly less uninviting in its physiognomy.

In 1602 the harbor of Catalina was visited by Viscaino, who reported it to be thickly inhabited by a people very ingenious, particularly in pilfering and

hiding their thefts. A priest who formed one of his party afterward described a kind of temple to the sun, with images and idols carved in soap-stone, which he asserted to exist in the vicinity of the harbor. The islands were doubtless inhabited down to the early portion of this century. Strange to say, those which are at present the most barren and desolate had once the largest population in proportion to their size.

In the early part of this century, the Russians, assisted by some American traders, extended their trading voyages to the coast of California and to the islands. While definite information seems unattainable, the traditions of the locality are to the following effect: About 1811, a Russian sea-otter party incurred hostilities from the natives of the islands, which resulted in a visit from a Russian cruiser, which attacked the natives, killing a number of them, and producing such consternation that the survivors returned to the main-land, and never returned. For nearly forty years the islands remained unvisited, when some persons who had conceived the idea of putting sheep upon them examined them with that project in view.

On San Miguel, so it is stated, they found one old native woman, who had dragged out a wretched existence in solitude since the affray with the Russians. She was taken to the main-land, and died there about 1864, the last of her race. After learning to communicate with the settlers, she informed them that she was quite a young girl when the islands were depopulated, and in the hasty departure of the survivors she had been left behind, probably by accident. Here she had lived alone, for nearly forty years.

There is nothing to confirm the existence of temples or idols, such as were described by Padre de la Ascencion; but of an abundant population, of great inventive ingenuity, and of different con-

ditions of climate and soil from those which now obtain, there are indisputable proofs. Near every ravine (including many now entirely dry, yet where water must formerly have been obtainable) the evidences of occupation, in the shape of shell-heaps, are conspicuous. Their magnitude and extent bear a direct relation to the abundance of water and the presence of a suitable landing-place for the canoes. Often they are of surprising thickness, indicating a prolonged occupation by the aborigines. The largest on Catalina Island are near the well-known springs, about five miles from the harbor.

Most of the islands are abundantly supplied with these shell-heaps, but they are particularly large and prolific in articles of use or ornament, on the outer desert islands of Clemente, San Nicolas, and on San Miguel. Those on Catalina are comparatively destitute of implements or human remains, and no exploration of them has revealed any signs of stone images or idols.

Various persons have visited these deposits, and returned more or less repaid with ethnological material—especially some of the officers of the U. S. Coast Survey, who so employed the leisure afforded by weather unfavorable for surveying work. Of these, by far the richest collection was made by Mr. Harford. It is now in the National Museum at Washington, and was obtained on the island of San Miguel. On visiting the shell-heaps on the north end of this island, the investigator is obliged to traverse about four miles of barren sandy wastes, which are intersected by sharp and broken ridges trending in an east and west direction. The shifting sands are raised in an unintermitting steady stream by the prevalent north-west trade-wind, and cut the face and blind the eyes like a shower of needles. Through the sand here and there may be seen scant spears of a slender, dry

grass; and in occasional patches the dry and dead remains of some shrubby plant, which in more favored localities raises a disproportionately small amount of tops compared with its knobby and swollen roots. Near the shell-heaps is a small grove of malva-trees, whose green leaves and penciled blossoms refresh the eyes. There are no young trees, however, as the omnipresent sheep crop every green thing within their reach close to the ground.

Arrived on the ground, we see, as we do everywhere else, that the surface of shifting sand is underlaid by a layer of vegetable mold, several feet in thickness, once soft and verdant, now baked and desiccated by the sun and wind. At one time this was evidently bare; since then, the wind has brought the sand in constantly increasing heaps, until the reaction began to set in, and the winds at present seem to tend slightly toward removing some of the sand. The north point is T-shaped, forming two small coves, in one or the other of which a boat can land in almost any weather. The point is bold and rather high, with small sandy beaches in the coves. On the edge of this bank or bluff lived our ancient people. Here we see, where the sea has worn it away, a shower of pearly shells from top to bottom of the bank; destitute, it is true, of the strength and cohesiveness which distinguishes them when fresh, but gaining rather than losing in their brilliant lustre by the very change which is destroying them. We find almost all the different kinds of abalone, limpets, and spiral pearly shells, which are still abundant in these waters. Though these cover the whole face of the bluff, yet they are derived from one stratum only. This varies from two or three to twenty feet in thickness, and is in great part identical with the stratum of soil which underlies the sand. The shell-layers extend for miles along the shore. When these banks were the

home of a myriad of busy and ingenious workmen, the island would seem to have been covered with verdure, and, in accordance with the climatic law, "to him that hath shall be given," rain must have been more abundant, and many springs have existed which are now dry. That these people cultivated the soil, there would seem to be little doubt. One of the most abundant relics to be found is the mortar and pestle, almost identical with those still used by the natives of New Mexico and Arizona for roughly grinding the seeds of grasses into a kind of flour.

Their houses were probably slight structures, which have disappeared and left no trace. Of their clothing we can form no idea from the record of the shell-heaps, but it is not improbable that it consisted of skins and woven grasses or vegetable fibres. Their food consisted of some cultivated vegetables or seeds, native fruits, fish, the flesh of seals and sea-lions whose bones abound in the shell-heaps, the smaller species of *cetacea*, birds, and particularly of the indigenous shell-fish, whose remains are everywhere before our eyes. Their means of navigation were probably canoes, made of materials derived from the main-land; perhaps, as among the Asiatic Esquimaux, obtained by barter. The selection of the outermost islands, by preference, as a place of residence, is accounted for by the greater abundance of shell-fish, otter, and marine animals, and by the greater protection afforded by their isolation from attacks from without. That these were a war-like people is evident, from the number of spear and lance heads found in the shell-heaps and graves. A doubt may be thrown on the courage of at least one of them, from the fact that among the articles obtained by Mr. Harford was a pelvic bone, with a long stone lance-head driven half through from behind; attesting at once the strength of arm of the

pursuer, and the fact that the defunct met his fate while running away. The majority of the weapons found are unmistakably for use in war, or in the chase; but there are certain heavy stone rings, or perforated balls, worked with a great deal of skill, and which have puzzled all the archæologists. The most probable explanation of the use of these gigantic beads, is that a knotted stick or thong was passed through the centre, and that they were then used as "slung-shots," or as the Patagonians use the *bolos*.

The activity of their domestic life is attested by the presence of fish-hooks made of abalone shell, stone spindles for twisting thread, the great variety of stone mortars and pestles (often with traces of rude ornamentation), and various nondescript but very curious bone tools and implements, one whole tool-chest of which we were fortunate enough to unearth. That they were not without relaxation may be inferred from the occurrence of sundry pipes, more like a modern cigar-holder in shape than anything else; and that their æsthetic tendencies were not wholly repressed, is shown by the discovery of an actual flute—a "tibia," in a double sense—made from the bone of an albatross, and pierced with five holes. This probably had a wooden mouth-piece originally, which had perished. Personal ornaments were in use, though mostly of a rather rude description. Beads and rough *wampum* strung on twisted sinew, alternating with sections of bones cut into "bugles," or with *hyqua*-shells, are very common. A touching record of parental love was disclosed in the myriads of beads and rounded plates of abalone pearl, which lay within a tiny baby's grave. The dead were laid in the earth near their habitations, apparently on their sides, with the knees drawn up toward the chin. With them are usually found implements and orna-

ments, once possessed by the dead, or contributed by the survivors as an equipment for him in the "happy hunting-grounds." One exception to this general rule may be noticed: On a sandy shelf of the bluff near the harbor a large number of interments have been made at a depth of a foot or two below the original surface. The drifting sands have increased this to about three feet. Here we unearthed some twenty skeletons, all of adults, and not a single implement or ornament, of native make, was found. One glass bead of red color, three-sided and of antique make, was found, showing that these natives had had, at least, a slight opportunity for intercourse with traders. It may be that these remains were the hastily interred bodies of those who fell in the last fight with the Russians, or were a body of the enemy cut off and destroyed in some internecine quarrel. It is but proper to say, however, that none of the bones showed evidences of bullet or lance wounds. In the other burial-grounds, remains are much more numerous. Mr. Harford in one place found seventy skulls within a short distance of each other. Many of them show the bone variously called the "interparietal," "supra-occipital," and "wormian bone," in the vicinity of the larger fontanel. This indicates that its presence can not be taken, as it has been by some ethnologists, as a proof of high antiquity or very degraded type. The *crania* are of medium size and development—*brachycephalic*; the teeth frequently showing evidences of caries existing before death, and always the flat-worn crowns, which indicate a diet of slightly cooked or gritty food.

These notes would be incomplete did they not include a mention of the peculiar effect of the drifting sands on exposed stones and bones. Nature's application of the sand-blast, recently patented for application in the mechanic arts, is very effectual. Bones are cut away, or

reduced to mere shells, in a short time; ordinary pebbles are cut in the direction of the prevailing winds into such fantastic forms that it is hardly possible to recognize them as natural productions. These curiously worn pebbles are found in New Zealand, on the coast of Oregon, and near the mouth of Klamath River, California, as well as on the islands; and specimens from all these places have been mistaken by sanguine collectors for articles of human handiwork. Apart from the dead shells exposed along the edges of the bluffs, the only means by which the collector may determine where to look for ethnological material, are the portions of mortars and other implements from which the wind has cleared away the sand. These exposed portions always show more or less evidence of the erosive action of sand carried by the wind, and some-

times are reduced to very small dimensions, in comparison with the parts which remain imbedded and thus protected.

Whether actively engaged in fishing or hunting—driving their light canoes with facile paddles; laboriously fashioning their weapons or implements of stone, bone, and pearly shell; or reposing in the golden glow of the western sun while they puffed the lazy wreaths of smoke from their stone pipes, pausing to catch the plaintive notes of the rude flute to which their children may have chanted, while the mother prepared the evening meal—the life of these ancient Lords of the Isles presents a pleasing contrast to that of the ordinary savage of to-day. Though they have passed away, “unwept, unhonored, and unsung,” yet the rude record of the shell-heap shall keep their memory green.

SOLUTION OF GEOLOGICAL AND ASTRONOMICAL PROBLEMS.

“That alone is true philosophy which repeats the words of the universe with fidelity, and is written as it were by dictation of the universe.”—*Bacon*.

GEOLGY establishes the fact beyond dispute, that the earth has repeatedly been the theatre of great physical activities, such as have resulted, successively, in radical changes throughout its entire surface. Continents have been changed in their configurations; mountains, shaken and rent, have had their fractured masses scattered over broad areas; forests have been buried, plains have taken the places of valleys, deserts of lakes; rivers have been altered in their courses and divided in their volumes; the distribution of climatic zones has been wholly changed—tropical succeeding to temperate, and temperate to frigid, in the same regions—and these in turn have passed away.

Various causes have been assigned for these phenomena. The most remarkable of them, however—such as are secondarily traceable to displacement of oceans—remain to be primarily accounted for. The writer offers a hypothetical solution of the problem they present, which he believes accords with philosophy and astronomical fact. A few introductory statements are necessary to a clear understanding of his idea.

Independent of the recognized motions of nutation and the precession of the equinoxes, all astronomers, from Hipparchus to those of the present period, have been cognizant of a slow polary movement of the earth, or gradual diminution of the obliquity of the ecliptic, progressing, according to Herschel, at the rate of 48" per century, or 1' in 125 years; a movement which is believed